Topology-Anchored, Context-Modulated MRI Segmentation with Deep Decoder Conditioning

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***Abstract*—Robust medical image segmentation under cross-scanner, cross-site, and appearance-driven distribution shift remains challenging because high-capacity encoders often entangle morphology with style, leading to unstable confidence and brittle predictions. We present a topology-anchored, context-modulated segmentation framework for magnetic resonance imaging (MRI) that preserves a compact implementation while explicitly separating structure-bearing latent information from residual appearance context. The proposed model starts from an nnU-Net-like encoder–decoder backbone and inserts selective Res-Mamba refinement at the bottleneck and deep decoder stage. From the shared bottleneck representation, a topology projection head produces a topology-focused latent code z_{topo}, while a residual context encoder extracts a complementary appearance/context code c_{app}. We then define a topology-anchored latent prior whose center is generated solely from z_{topo} and whose spread is modulated by both z_{topo} and c_{app}. The prior is converted into anchor tokens that condition the deep decoder by cross-attention at D3. Training is driven by three objectives only: a segmentation loss, a topology-aware metric loss that structures the latent space using topology/morphology-derived positive and negative relations, and an anchor regularization loss that constrains each topology latent to remain near a sample-adaptive topology anchor. At inference, the same anchor induces a Mahalanobis-type plausibility score that calibrates raw segmentation confidence. This manuscript details the architecture, mathematical formulation, training workflow, implementation blueprint, and evaluation protocol for the proposed framework in IEEE Transactions on Medical Imaging style.**

***Index Terms*—MRI segmentation, topology-aware representation learning, latent anchor modeling, decoder conditioning, calibration, medical image analysis.**

I. Introduction

MEDICAL image segmentation remains a central problem in MRI analysis, yet performance often degrades when appearance shifts across scanners, protocols, institutions, or patient populations. Contemporary encoder–decoder models can attain high in-distribution Dice scores while still overfitting to intensity style, texture statistics, and acquisition-specific artifacts. In practice, this makes the latent representation unstable under domain shift and weakens the reliability of confidence estimates.

Two families of methods are particularly relevant to this problem. First, strong biomedical segmentation backbones such as U-Net and nnU-Net establish highly competitive encoder–decoder baselines for many datasets and tasks [1], [2]. Second, recent sequence modeling modules such as Mamba provide efficient long-range context aggregation and can enrich deep feature refinement without fully replacing convolutional backbones [3]. However, a strong backbone and a strong decoder do not by themselves guarantee that the latent geometry will reflect stable anatomical structure rather than appearance-specific shortcuts.

The design goal of this work is therefore not only to improve segmentation accuracy but also to make the deep latent representation more structure-driven and more useful for plausibility estimation. Rather than requiring explicit topological descriptors at inference, we use topology- and morphology-derived supervision during training to shape a latent branch toward structure-consistent organization. We then anchor that latent branch with a sample-adaptive prior whose center is driven by topology and whose spread is modulated by residual context. This formulation yields a compact model with only three losses while preserving a clear conceptual separation between structure and residual appearance.

The main contributions of the proposed framework are fourfold. First, we introduce a topology-anchored latent prior whose mean is determined by a topology-focused latent code rather than by a generic context code. Second, we use a residual context branch only to modulate anchor spread, which prevents context from dominating anchor location. Third, we convert the latent prior into anchor tokens that condition the deep decoder through cross-attention. Fourth, we use the same prior to compute an inference-time Mahalanobis-type plausibility score for self-calibrated confidence.

II. Related Work

Encoder–decoder architectures remain the standard for biomedical segmentation. U-Net introduced the core strategy of combining multi-scale encoding with skip-connected decoding [1]. nnU-Net demonstrated that careful self-configuration of preprocessing, patch sizing, data augmentation, and training heuristics yields a highly competitive default pipeline across many biomedical benchmarks [2]. Our work retains the strong inductive bias of an nnU-Net-like hierarchy and extends it with selective sequence-modeling refinement and topology-anchored latent conditioning.

Representation learning methods provide a natural mechanism for structuring latent spaces. Supervised contrastive learning offers a principled way to pull semantically related samples together while pushing unrelated samples apart [4]. We adopt this philosophy but define relation sets using topology/morphology-derived supervision rather than generic class labels. This makes the latent space reflect structural similarity more directly.

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Confidence estimation and calibration are also closely related to the proposed approach. Classical calibration methods study the mismatch between predictive probabilities and empirical correctness [5]. Mahalanobis-based distances have also been used as a practical measure of feature plausibility and out-of-distribution behavior [6]. Our framework differs in that the plausibility score is not computed against a single global feature distribution; instead, each sample receives its own topology-anchored, context-modulated latent prior.

III. Method Overview

The proposed model consists of four parts: 1) an nnU-Net-like visual backbone with selective Res-Mamba refinement; 2) a topology projection head that maps the shared deep representation to a topology-focused latent code; 3) a residual context branch that captures non-topological variability; and 4) a topology-anchored, context-modulated anchor module whose tokens condition the deep decoder and whose distribution regularizes the latent space.

Fig. 1 shows the full framework. The model takes an input MRI image x, produces a shared bottleneck representation h, and then splits into a topology branch and a residual context branch. The topology branch generates z_{topo}, which drives the anchor center. The context branch generates c_{app}, which modulates the anchor spread. Together with z_{topo}, the resulting anchor parameters are converted into anchor tokens that condition the deep decoder via cross-attention at D3. During training, topology/morphology-derived relations from ground-truth masks supervise the latent branch through a metric objective. During inference, the same anchor provides a latent plausibility score for confidence calibration.

IV. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote the training set, where x_i is an MRI slice or volume and y_i is the corresponding segmentation mask. The segmentation network predicts a probability map p_i and the final hard segmentation $\hat{y}_i$ is obtained by thresholding in the binary case or by argmax in the multi-class case.

Our goal is to learn a model that simultaneously satisfies three requirements:

1) produces accurate segmentation masks,
2) organizes the latent representation according to topology/morphology-driven structural similarity, and
3) provides a sample-adaptive topology anchor that supports both decoder conditioning and confidence calibration.

To achieve this, we define a topology-focused latent variable z_{topo_i} and a residual context variable c_{app_i} from the shared representation h_i. The topology anchor center is generated from z_{topo_i}, whereas the anchor spread depends on both z_{topo_i} and c_{app_i}.

V. Visual Backbone and Shared Representation

The visual trunk follows an nnU-Net-like hierarchy with four encoder stages and a mirrored decoder [2]. Each encoder stage consists of convolutional blocks with downsampling, while each decoder stage fuses an upsampled representation with its corresponding skip feature. We preserve this strong baseline because it provides robust multi-scale anatomical representation learning.

To improve long-range structural modeling without over-complicating the network, we insert selective Res-Mamba refinement at two locations only: the bottleneck after E4 and the deep decoder stage D3. Let $E_\theta(\cdot)$ denote the encoder and bottleneck computation. For sample i,

$$h_i = E_\theta(x_i). \tag{1}$$

Here, h_i is the shared deep representation that already contains rich spatial semantics, but it still entangles topology, appearance, and residual context. The rest of the method separates these factors at the bottleneck rather than forcing one latent code to carry all of them.

VI. Topology Projection Head

The topology projection head is deliberately lightweight. It should compress the shared feature into a compact structural code rather than act as a second backbone. Let $h_i \in \mathbb{R}^{C\times H'\times W'}$ for 2-D MRI or $h_i \in \mathbb{R}^{C\times H'\times W'\times D'}$ for 3-D MRI. We first apply global pooling to remove spatial resolution while preserving global structural evidence:

$$u_i^{\text{topo}} = \text{Concat}(\text{GAP}(h_i), \text{GMP}(h_i)). \tag{2}$$

We then map the pooled vector to the topology latent code through a small multi-layer perceptron:

$$\tilde{u}_i = W_1^{\text{topo}} u_i^{\text{topo}} + b_1^{\text{topo}}, \tag{3}$$

$$\hat{u}_i = \text{GELU}(\text{LN}(\tilde{u}_i)), \tag{4}$$

$$z_{\text{topo}_i} = W_2^{\text{topo}} \hat{u}_i + b_2^{\text{topo}}. \tag{5}$$

Optionally, z_{topo_i} may be L2-normalized when cosine similarity is used inside the metric loss. The key point is that the projection head does not know topology a priori; it becomes topology-focused because the downstream objectives repeatedly reward structural consistency and punish topology-inconsistent embeddings.

VII. Residual Context Encoder

The residual context encoder is intentionally separated from the topology branch. Its purpose is not to determine where the latent anchor is centered, but rather to model variability that modulates the uncertainty or spread of the anchor. From the same shared representation h_i, we compute a pooled summary

$$u_i^{\text{app}} = \text{Concat}(\text{GAP}(h_i), \text{GMP}(h_i)), \tag{6}$$

and map it to a residual context code

$$c_{\text{app}_i} = C_\xi(u_i^{\text{app}}). \tag{7}$$

The context code absorbs factors such as scanner style, acquisition noise, and residual appearance statistics that should not control anchor location but may reasonably affect how tightly the topology latent should be constrained.

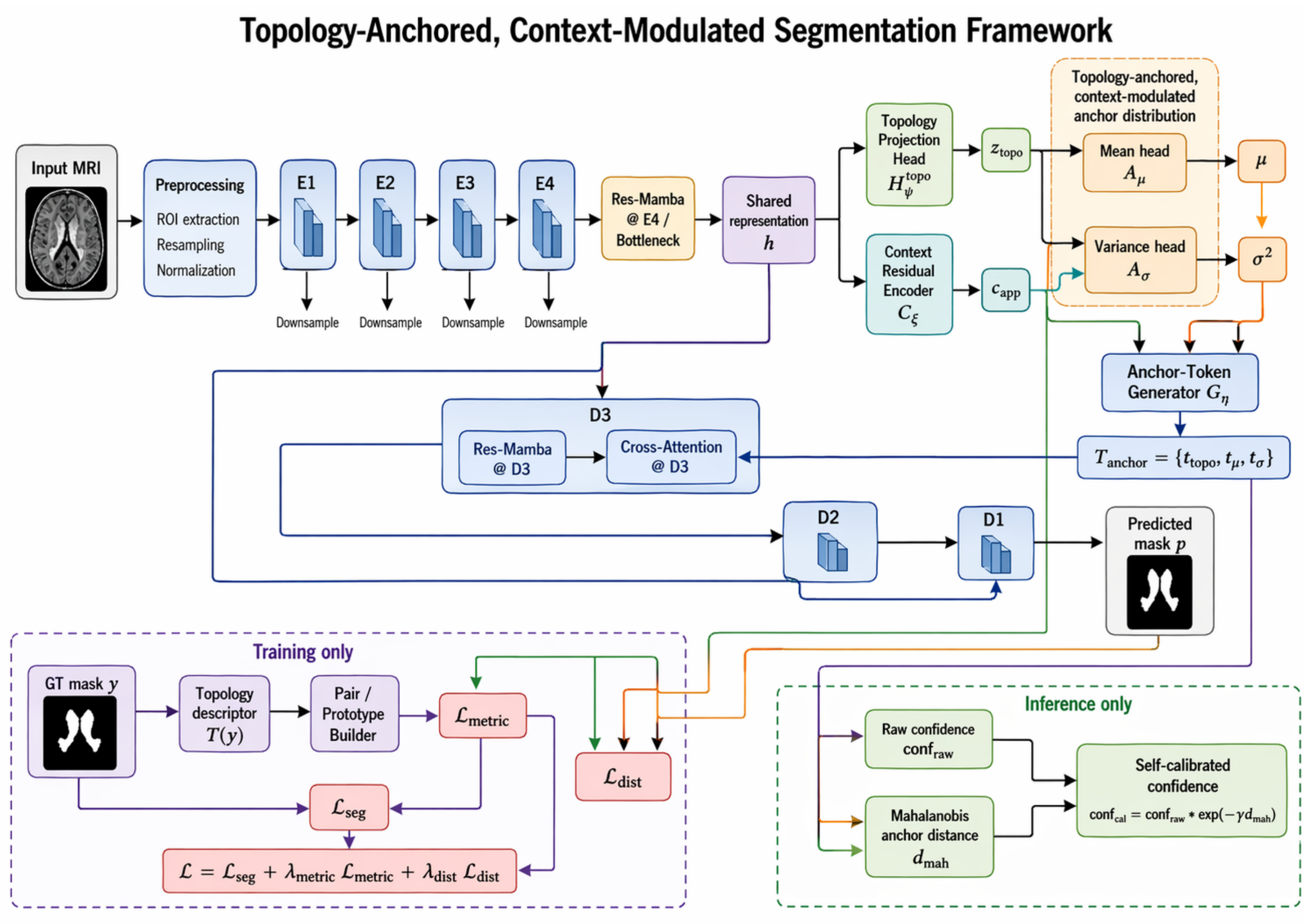

Fig. 1. Topology-anchored, context-modulated segmentation framework. The shared bottleneck feature h is split into a topology latent branch and a residual context branch. The anchor center is generated from z_{topo}, while the anchor spread is generated from z_{topo} and c_{app}. Anchor tokens condition the deep decoder at D3. Training uses three losses only: segmentation, topology-aware metric structuring, and topology-anchor regularization. Inference uses the same anchor to compute a Mahalanobis-type plausibility score for self-calibrated confidence.

VIII. Topology-Anchored, Context-Modulated Anchor Distribution

The anchor module is the central conceptual change relative to a generic context-anchored design. We define the anchor center using the topology latent only:

$$\mu_i = A_\mu(z_{\text{topo}_i}), \tag{8}$$

where A_μ is a lightweight mean head. This ensures that anchor position is topology-driven. We then generate the diagonal variance using both topology and residual context:

$$\sigma_i^2 = A_\sigma(z_{\text{topo}_i}, c_{\text{app}_i}), \tag{9}$$

with positivity enforced through a softplus or exponential transform. The resulting latent prior is

$$p_i(z) = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2)). \tag{10}$$

This distribution has a clear interpretation: topology decides where the sample should sit in latent space, while residual context decides how much deviation is plausible.

IX. Anchor-Token Generator and Decoder Conditioning

The anchor tokens transform the topology latent and the anchor parameters into decoder-readable conditioning vectors. We define

$$T_{\text{anchor}\,i} = G_\eta(z_{\text{topo}_i}, \mu_i, \sigma_i^2) = \{t_{\text{topo}}^{(i)}, t_\mu^{(i)}, t_\sigma^{(i)}\}. \tag{11}$$

These tokens are injected at D3, the deepest decoder stage that still has enough semantic abstraction to benefit from structural conditioning. Let $f_{D3,i}^{\text{rm}}$ denote the D3 feature after skip fusion and Res-Mamba refinement. We then form query, key, and value projections as

$$Q_i = W_Q f_{D3,i}^{\text{rm}}, \tag{12}$$

$$K_i = W_K T_{\text{anchor}\,i}, \tag{13}$$

$$V_i = W_V T_{\text{anchor}\,i}. \tag{14}$$

Cross-attention is computed as

$$\text{Attn}_i = \text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i, \tag{15}$$

and the conditioned decoder feature is

$$\tilde{f}_{D3,i} = f_{D3,i}^{\text{rm}} + \text{Attn}_i. \tag{16}$$

The remainder of the decoder produces the segmentation probability map

$$p_i = D_\phi(h_i, T_{\text{anchor}i}). \tag{17}$$

The conditioning therefore occurs neither at the input nor at all scales, but at a deep decoder stage where global structural guidance is most useful and least redundant.

X. Topology Supervision and Pair Construction

Topology is supplied during training only. Given a ground-truth mask y_i, we compute a topology/morphology descriptor

$$s_i = T(y_i), \tag{18}$$

where $T(\cdot)$ may include connected-component statistics, ring- or cavity-aware morphology cues, Betti-like summaries, or persistence-inspired shape descriptors, depending on the chosen implementation. These descriptors are not required during inference.

A pair/prototype builder uses the descriptor set $\{s_i\}$ within a batch to determine which samples are structurally similar enough to be treated as positives and which should be treated as negatives. For anchor i, let $P(i)$ denote the positive set. The latent branch is then supervised through a topology-aware metric objective that uses these sets rather than relying on raw appearance similarity.

XI. Training Objectives

We intentionally retain only three losses in order to keep optimization stable and the contribution easy to interpret.

A. Segmentation Loss

The segmentation loss combines Dice overlap with cross-entropy or binary cross-entropy:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{seg}}^{\text{Dice}}(p, y) + \mathcal{L}_{\text{seg}}^{\text{CE/BCE}}(p, y). \tag{19}$$

This term preserves the core segmentation objective and ensures that all latent conditioning remains useful for the end task.

B. Topology-Aware Metric Loss

The metric loss is applied to the topology latent z_{topo} rather than to a generic latent code. Using supervised contrastive learning [4], we write for anchor i

$$\mathcal{L}_{\text{metric}}^{(i)} = \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_{\text{topo}_i}, z_{\text{topo}_p})/\tau)}{\sum_{a \neq i} \exp(\text{sim}(z_{\text{topo}_i}, z_{\text{topo}_a})/\tau)}. \tag{20}$$

The batch metric loss is the mean over anchors. This loss is the main force that makes the topology projection head discover structure-consistent features from the shared representation.

C. Topology-Anchor Distribution Loss

For each sample, the topology latent is regularized toward its own topology anchor:

$$\mathcal{L}_{\text{dist}}^{(i)} = \frac{1}{2} \sum_j \left[\frac{(z_{\text{topo}i,j} - \mu_{i,j})^2}{\sigma_{i,j}^2 + \epsilon} + \log(\sigma_{i,j}^2 + \epsilon) \right]. \tag{21}$$

This term has two effects. First, it keeps the latent representation from drifting arbitrarily once metric clusters are formed. Second, it makes the anchor parameters meaningful because they must explain the latent position of each sample. The total training objective is therefore

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{metric}} \mathcal{L}_{\text{metric}} + \lambda_{\text{dist}} \mathcal{L}_{\text{dist}}. \tag{22}$$

XII. Why the Projection Head Learns Topology-Relevant Features

The projection head does not compute topological descriptors explicitly from the image. Instead, it learns a feature selection rule through backpropagation. During early training, the output of H_ψ^{topo} is likely to mix structure, style, and noise. However, if two samples have the same structural class but different scanner appearance, the metric loss still treats them as positives. Any feature that keeps them far apart will be penalized, while any feature that makes them close will be rewarded. Repeating this process over many batches gradually suppresses appearance-only cues and amplifies structure-consistent cues.

The anchor regularization then adds a second pressure. Once a topology-consistent cluster begins to form, the anchor center predicted from z_{topo} becomes a local attractor for samples in that cluster. The variance head uses residual context to decide whether the attractor should be tight or loose. Thus the final topology latent is not simply a contrastive embedding; it is also a topology-anchored embedding whose dispersion depends on residual uncertainty. In short, topology emerges as the most efficient latent explanation for simultaneously minimizing the metric loss and the anchor regularization loss.

XIII. Inference and Self-Calibrated Confidence

Inference requires only the image x. The model computes

$$x \to h \to z_{\text{topo}}, c_{\text{app}} \to \mu, \sigma^2 \to T_{\text{anchor}} \to p \to \hat{y}. \tag{23}$$

The raw segmentation confidence conf_{raw} can be obtained from the predicted probability map, for example by averaging class confidence over the predicted foreground. To estimate latent plausibility, we compute a Mahalanobis-type anchor distance,

$$d_{\text{mah}} = \sum_j \frac{(z_{\text{topo}_j} - \mu_j)^2}{\sigma_j^2 + \epsilon}. \tag{24}$$

A self-calibrated confidence is then defined as

$$\text{conf}_{\text{cal}} = \text{conf}_{\text{raw}} \exp(-\gamma d_{\text{mah}}), \tag{25}$$

where $\gamma > 0$ controls how strongly topology-anchor plausibility suppresses overconfidence. This yields an interpretable confidence mechanism: if the segmentation head is confident but the topology latent is implausible under its own anchor, the final confidence is reduced.

XIV. Implementation Blueprint

For a professional project implementation, we recommend the following modular decomposition. The backbone, decoder, topology projection head, residual context encoder, anchor heads, and token generator should live in separate source files under a `models/` package. Losses should be implemented in a `losses/` package with separate files for segmentation loss, topology-aware metric loss, and anchor regularization. Topology descriptor extraction, pair construction, and latent-space diagnostics should be placed under a `topology/` package. Dataset manifests, transforms, preprocessing scripts, and data verification utilities should be grouped under `data/`. Evaluation, calibration, plotting, and visualization routines should be isolated under `evaluation/` and `visualize/`. Hyperparameters should be controlled through configuration files rather than hard-coded constants.

A practical training loop for each minibatch proceeds as follows. First, load (x_i, y_i) and apply preprocessing and data augmentation. Second, compute h_i, then split it into z_{topo_i} and c_{app_i}. Third, generate μ_i, σ_i^2, and $T_{\mathrm{anchor}i}$. Fourth, decode to obtain p_i. Fifth, construct topology/morphology relation sets from y_i. Sixth, compute $\mathcal{L}_{\mathrm{seg}}$, $\mathcal{L}_{\mathrm{metric}}$, and $\mathcal{L}_{\mathrm{dist}}$, sum them, and update all trainable parameters with a single optimizer step.

XV. Experimental Protocol Template

This manuscript focuses on method and architecture. A complete TMI submission should add the dataset-specific experimental sections once the implementation is finalized. Recommended evaluation should include: 1) standard segmentation quality metrics such as Dice, Jaccard, Hausdorff distance, and surface distance; 2) in-distribution versus distribution-shift evaluation across scanners, sites, or vendors; 3) confidence calibration metrics such as expected calibration error and risk–coverage curves; and 4) ablations isolating the contribution of the topology latent branch, topology anchoring, context modulation, and deep decoder cross-attention.

For clarity and scientific rigor, ablations should compare at least the following variants: a plain nnU-Net-like baseline, the backbone with selective Res-Mamba only, the backbone plus topology-aware metric learning only, the context-anchored variant, and the final topology-anchored, context-modulated variant. Additional visualizations should include latent-space plots, anchor-center trajectories, confidence histograms, and failure cases under scanner shift.

XVI. Discussion

The proposed design is intentionally balanced. It is more structured than a generic latent regularization approach because the anchor center is explicitly tied to the topology latent. At the same time, it is lighter and easier to optimize than a fully explicit topology-prediction framework with many auxiliary heads and losses. This balance is especially attractive when the research goal is to demonstrate topology-informed robustness and calibration without turning the model into a topological inference engine.

The main limitation is that topology supervision remains descriptor-driven rather than formally guaranteed. The learned latent code is topology-aware in the sense that it is shaped by topology/morphology-derived relations and anchored accordingly, but it is not a formal vector of Betti numbers or persistent homology coordinates. If stronger topological guarantees are desired, future work may add an explicit topology prediction branch or a differentiable topology-consistency term on the predicted masks.

XVII. Conclusion

We presented a topology-anchored, context-modulated MRI segmentation framework that preserves the practical strengths of an nnU-Net-like backbone while restructuring the latent space around topology-consistent anchors. A topology projection head produces a structure-focused latent embedding; a residual context encoder models non-topological variation; the anchor center is generated solely from the topology latent; and the anchor spread is modulated by topology and residual context. The resulting anchor tokens condition the deep decoder, while the same latent anchor supports plausibility-aware confidence calibration. The full method is trained with three losses only: segmentation, topology-aware metric structuring, and topology-anchor regularization. This yields a methodologically clean and implementation-ready foundation for a robust TMI-style study of structure-aware MRI segmentation.

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